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Source: *The American Mathematical Monthly*, Mar., 2004, Vol. 111, No. 3 (Mar., 2004), pp. 251-255

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A Note on Isometric Embeddings of Surfaces of Revolution

Martin Engman

1. INTRODUCTION. Among the first, and most concrete, examples of Riemannian manifolds encountered by a student of mathematics are simple surfaces, such as spheres and tori, that we can "see" because they are "sitting inside" three-dimensional Euclidean space. Soon after the abstract definition of Riemannian manifold was formulated, the question arose: Can all Riemannian manifolds be realized, while preserving distances, as submanifolds "sitting inside" a sufficiently large dimensional Euclidean space? In other words: Can Riemannian manifolds be isometrically embedded in $\mathbb{R}^n$ for large enough n? A very general answer to this question was provided by Nash in 1956 [**7**]. This result receives some attention in the book (and subsequent film) *A Beautiful Mind* [**8**], albeit in a supporting role to game theory's leading role.

In 1916 Weyl tackled the following special case: Does there exist a smooth global isometric embedding of the Riemannian manifold (S^2, g) into $(\mathbb{R}^3, can)$? Here g is a given metric on the sphere and can is the flat (i.e., canonical) metric on $\mathbb{R}^3$. There are many solutions to this problem under the added assumption that the Gauss curvature is positive (see, for example, [**11**], [**9**], [**4**], [**1**], and [**10**]) or at least nonnegative (see [**5**]). The converses of these theorems cannot hold, for it is easy to find embedded metrics with some negative curvature. Moreover, not every metric on S^2 admits such an isometric embedding (see [**3**]).

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In this paper we study the special case of S^1-invariant metrics on S^2. These metrics have two distinguished points, the north and south poles, about which we can center disks of constant geodesic radii. By restricting ourselves to embeddings whose images are classical surfaces of revolution, we are able to give an elementary proof, accessible to calculus students, that the nonnegativity of the integrals of the curvature over all disks centered at a pole is the necessary and sufficient condition for isometric embeddability as a classical surface of revolution. Thus, although an isometrically embedded metric cannot in general be expected to have nonnegative curvature everywhere, in some sense we have, in this special case, the next best thing: every pole-centered disk has nonnegative integral curvature.

2. ISOMETRIC EMBEDDINGS OF ROTATION INVARIANT SURFACES. We say that a Riemannian manifold (M, g) (in our case, a surface) is S^1-*invariant* if its isometry group (i.e., group of distance preserving diffeomorphisms from M to itself) contains the circle S^1 as a subgroup. The results that we obtain are based on a well-known and quite elementary result regarding S^1-invariant metrics on S^2 that can be found in Besse [**2**, pp. 95–106]. The reader familiar with Besse's book will find the results of this section to be, essentially, nothing but a reformulation of his treatment. We use the terminology *surface of revolution* when referring to a classical surface of revolution, i.e., to a surface generated by the rotation of a curve about a line. This is an example of an *equivariant embedding*. An S^1-invariant metric on S^2 is sometimes called an *abstract surface of revolution* (see the definition in Hwang [**6**]).

We henceforth assume that $\mathbb{R}^3$ is endowed with its standard flat metric and therefore suppress any further mention of its metric. Let (M, g) be an S^1-invariant Riemannian manifold that is diffeomorphic to S^2. We assume the metric to be of class C^∞. There are exactly two points that are fixed by the action of the effective S^1 isometry group. We call these fixed points np and sp and use U to denote the open set $M \setminus \{np, sp\}$. On U the metric has the form

$$g = ds \otimes ds + a^2(s) d\theta \otimes d\theta, \tag{1}$$

where s is the arclength along a geodesic connecting np to sp, L is the length of such a geodesic, and $a : [0, L] \to \mathbb{R}^+$ is a function satisfying $a(0) = a(L) = 0$ and $a'(0) = 1 = -a'(L)$. The canonical (i.e., constant curvature) metric is obtained by taking $a(s) = \sin s$.

The Riemannian measure $\sqrt{\det g}\, ds \wedge d\theta$ (with respect to which all surface integrals are computed as double integrals) is, in this case, given by $dA = a(s) ds \wedge d\theta$. This fact, together with the following formula for the Gauss curvature of this metric,

$$K(s) = -a''(s)/a(s), \tag{2}$$

reduces the proof of the Gauss-Bonnet theorem, which asserts that $\int_M K = 4\pi$, to a simple application of the fundamental theorem of calculus.

One can always isometrically embed such a metric into $\mathbb{R}^2 \times \mathbb{C}$ with a mapping $\Psi = (\psi^1, \psi^2, \psi^3)$ defined as follows:

$$\begin{cases} \psi^1(s, \theta) = a(s) \cos\theta, \\ \psi^2(s, \theta) = a(s) \sin\theta, \\ \psi^3(s, \theta) = \int_c^s \sqrt{1 - (a'(t))^2}\, dt, \end{cases} \tag{3}$$

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for some c in $[0, L]$. It is easy to see that the function ψ^3 is real-valued if and only if $|a'(s)| \leq 1$ and, in this case, (3) clearly provides an embedding as a surface of revolution revolved about the z-axis (= the ψ^3-axis). This gives a heuristic argument in support of the following proposition. For detailed proofs see [**2**] or the generalization given by Hwang in [**6**].

Proposition 1. *Let (M, g), with metric g as in (1), be diffeomorphic to S^2. Then (M, g) admits an isometric C^1-embedding in $\mathbb{R}^3$ as a surface of revolution if and only if $|a'(s)| \leq 1$ for all s in $[0, L]$.*

We end this section with the observation that, in the special case under consideration, nonnegative curvature implies isometric embeddability as a surface of revolution: $K(s) \geq 0$ and (2) imply that $a'(s)$ is a nonincreasing function on $[0, L]$ with maximum value 1 and minimum -1, whence $|a'(s)| \leq 1$. This establishes the following proposition, which can be thought of as a special case of the theorem of Hong and Zuily [**5**].

Proposition 2. *Let (M, g) be an S^1-invariant Riemannian manifold that is diffeomorphic to S^2. If its Gauss curvature K has the property that $K \geq 0$ on M, then (M, g) has an isometric C^1-embedding in $\mathbb{R}^3$ as a surface of revolution.*

3. MAIN RESULTS. Our main result is the following necessary and sufficient condition.

Theorem 3. *Let (M, g) be an S^1-invariant Riemannian manifold that is diffeomorphic to S^2. Let np and sp denote the fixed points of the S^1-action, and let K be the Gauss curvature of (M, g). Then (M, g) has an isometric C^1-embedding in $\mathbb{R}^3$ as a surface of revolution if and only if*

$$\int_\Omega K \geq 0$$

for all geodesic disks Ω centered at np or sp.

Proof. As in Figure 1, one can think of the pole-centered disks as "polar caps." Each closed disk Ω centered at np is parametrized by $0 \leq s \leq x$ and $0 \leq \theta < 2\pi$ for some x. Then

$$\int_\Omega K = \int_0^{2\pi} \int_0^x K(s)a(s)\, ds \wedge d\theta = 2\pi \int_0^x K(s)a(s)\, ds.$$

Hence $\int_\Omega K \geq 0$ if and only if $\int_0^x K(s)a(s)\, ds \geq 0$. But, since $K(s) = -a''(s)/a(s)$ and $a'(0) = 1$,

$$\int_0^x K(s)a(s)\, ds = -a'(x) + 1, \tag{4}$$

and it follows easily that $\int_0^x K(s)a(s)\, ds \geq 0$ if and only if $a'(x) \leq 1$.

A similar argument shows that the nonnegativity of all integrals over disks centered at sp is equivalent to $\int_x^L K(s)a(s)\, ds \geq 0$ for all x. Also, it is easy to show that

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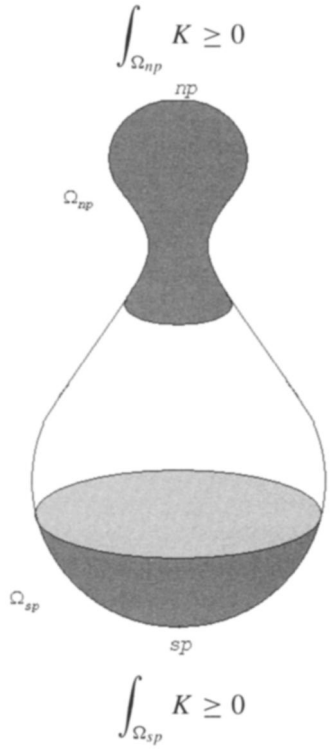

Figure 1. Nonnegative integral curvature on the polar caps Ω_{np} and Ω_{sp}.

$\int_0^L K(s)a(s)\,ds = 2$ (this is the Gauss-Bonnet theorem), hence that

$$\int_x^L K(s)a(s)\,ds = 2 - \int_0^x K(s)a(s)\,ds.$$

After substitution of (4), this leads to the conclusion that $\int_x^L K(s)a(s)\,ds \geq 0$ if and only if $a'(x) \geq -1$. In other words, $\int_\Omega K \geq 0$ for all disks Ω centered at np or sp if and only if $|a'(s)| \leq 1$ for all s in $[0, L]$. Theorem 3 now follows from Proposition 1. ■

There are several interesting consequences of this result. Perhaps the first observation to make is that Theorem 3 provides another proof of Proposition 2 because $K \geq 0$ everywhere on M obviously implies that $\int_\Omega K \geq 0$ for all Ω.

It is an elementary fact that the poles of a classical surface of revolution diffeomorphic to S^2 are umbilics and hence are points of nonnegative curvature. Turning Theorem 3 around yields a slightly more abstract version of this result that can be thought of as an obstruction in terms of curvature to embedding a manifold in $\mathbb{R}^3$ as a surface of revolution.

Corollary 4. *Let (M, g) be an S^1-invariant Riemannian manifold that is diffeomorphic to S^2. Let np and sp denote the fixed points of the S^1-action, and let K be the Gauss curvature of (M, g). If $K(np) < 0$ or $K(sp) < 0$, then (M, g) cannot be isometrically embedded in $\mathbb{R}^3$ as a surface of revolution.*

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Proof. We assume $K(np) < 0$, i.e., $K(0) < 0$. Since $K(s)$ extends to a continuous function on $[0, L]$ and $a(s) > 0$ on $(0, L)$, for some $\epsilon > 0$ we have $K(s)a(s) < 0$ on $(0, \epsilon)$ and hence $\int_0^\epsilon K(s)a(s)ds < 0$. The result now follows from Theorem 3. ■

Another particular case of the Gauss-Bonnet theorem states that for a simply connected region $\Omega \subset M$ with C^2-boundary C

$$\int_\Omega K + \oint_C k_g = 2\pi, \tag{5}$$

where k_g is the geodesic curvature of C. Applying this to geodesic disks centered at a pole and observing that the boundary of any such disk is a circle of latitude, we get:

Corollary 5. *Let (M, g) be an S^1-invariant Riemannian manifold that is diffeomorphic to S^2 and has Gauss curvature K. Then (M, g) has an isometric C^1-embedding in $\mathbb{R}^3$ as a surface of revolution if and only if for all circles of latitude C it is the case that*

$$\left|\oint_C k_g\right| \leq 2\pi.$$

Proof. Each circle of latitude C serves as the boundary of two disks, Ω_1 centered at np and Ω_2 centered at sp, that impose opposite orientations on C. From (5) we infer

$$\int_{\Omega_1} K + \oint_C k_g = 2\pi, \quad \int_{\Omega_2} K - \oint_C k_g = 2\pi.$$

Thus $\int_\Omega K \geq 0$ for all "polar caps" Ω if and only if $2\pi - \oint_C k_g \geq 0$ and $2\pi + \oint_C k_g \geq 0$, where C is the boundary circle of Ω. This translates to the requirement that $\left|\oint_C k_g\right| \leq 2\pi$ for all circles of latitude C. ■

ACKNOWLEDGMENTS. The author thanks Gus Engman for Figure 1 and J. J. Duistermaat for pointing out errors in the preliminary version of this note. This work was partially supported by the NSF Grant: Model Institutes for Excellence at UMET.

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